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Nanoscale banded 90° domain patterns in (111) oriented $(Pb,La)TiO_3$ thin films

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Lanthanum doped lead titanate ferroelectric thin films were grown on $Pt/Ti/SiO_2/Si(100)$ substrates using rf magnetron sputtering. X-ray diffraction patterns show that the main phase of the films is tetragonal perovskite with preferred orientation of (111) perpendicular to the substrate. We investigated the ferroelectric domain patterns and the corresponding polarization arrangements of the films using piezoresponse force microscopy. Head-to-tail polarization configurations were often found to exist, indicating that the low electrical energy configurations are quite common in these thin films. Nanoscale banded 90° *a-a* domain patterns as small as 30–60 nm in width were observed. The banded domains alternated with adjacent lamellae can be attributed to mechanical strains at the surface and at the interface between the substrate and the film. © *2006 American Institute of Physics*. [DOI: 10.1063/1.2357865]

I. INTRODUCTION

Great attention has been paid to preparations, characterizations, and applications of ferroelectric thin films in the past decades. Because of their unique pyroelectric and ferroelectric properties, $(Pb_xLa_{1-x})Ti_{1-x/4}O_3$ (PLT) thin films have been considered as one of the most promising materials for room-temperature pyroelectric infrared sensors, infrared thermal imaging devices,[1,2] etc.

Determining the domain patterns in ferroelectric films is one of the key requisites for a better understanding of their piezoelectric, pyroelectric, and ferroelectric properties.[3–5] In addition, the switching of polarization of PLT films is accompanied by a mechanical piezoelectric effect which indicates spatial reorganization of domains. A piezoelectric vibration of the PLT films on a Pt electrode can be induced by applying an electric field of certain frequency (9 kHz) to the electrode. The measurement of the vibration amplitude on the film surface allows one to assign the polarization state and direction of ferroelectric films by piezoresponse force microscopy (PFM).[6]

Studies on the local ferroelectric properties of a film at the nanoscale will provide an effective way not only for achieving insights into fundamental issues such as polarization reversal, domain switching, and domain structure but also for optimizing the properties of the film.[7] In this paper, PLT thin films were grown on $Pt/Ti/SiO_2/Si(100)$ substrates using rf magnetron sputtering. The crystalline and domain properties were studied by x-ray diffraction (XRD) and PFM.

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II. EXPERIMENT

A conventional process for mixing oxide was used to prepare the lanthanum doped lead titanate ceramic targets for sputtering. Ceramics with the nominal composition $(Pb_{0.9}La_{0.1})Ti_{0.975}O_3$ (PLT) were synthesized, and excessive 10 mol % PbO was added to compensate the volatilization loss of lead in the process of target preparation. The starting materials used in this study were La_2O_3, TiO_2, and PbO with purities of 99.99%, 99.0%, and 99.0%, respectively. They were milled in a polyethylene jar with agate balls for 24 h using ethanol as a medium, dried and calcined at 850 °C for 2 h. The calcined powders were milled into fine powder again, and then poly(vinylalcohol) was added as a binder. The powders with binder were subsequently granulated and pressed into disks of 70 mm diameter and 10 mm thickness under 40–60 MPa uniaxial pressure and sintered at 1000 °C for 2 h to form the PLT ceramic targets.

The PLT ferroelectric thin films with thickness of 1 μm were fabricated on $Pt/Ti/SiO_2/Si(100)$ substrates by using rf magnetron sputtering system. For the sake of good compatibility with planar technology of semiconductors and avoiding volatilization loss of lead, the substrate temperature could not be too high during deposition. Hence, we first deposited films at substrate temperature of 300 °C and at a pressure of 2 Pa by using pure argon and annealed them at various temperatures *ex situ* by using conventional thermal annealing (CTA) processing in a lead compensated atmosphere to achieve desired crystallinity. The heating rate from room temperature to a desired temperature is 3 °C/min and the sintering time is 1 h. Typical deposition conditions are shown in Table I.

The crystallographic analysis of the PLT thin films has been carried out by x-ray diffraction (DX-1000). The topography and domain patterns of the PLT thin films were obtained by means of a scanning force microscope (SFM)

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TABLE I. Typical deposition conditions of the PLT thin films.

Item	Parameter
Target	$Pb_{0.9}La_{0.1}TiO_3$ ceramics
Substrate	$Pt/Ti/SiO_2/Si(100)$
Target-substrate distance	40 mm
Substrate temperature	300 °C
Working gas composition	9:1 ($Ar:O_2$)
Working pressure	2 Pa
rf input power density	2.5 W/cm^2
Deposition time	3 h
Deposition rate	0.9 Å/s

(Seiko SPA 300) in piezoresponse mode, i.e., PFM, in which an ac voltage $V_{ac}=V_0 \sin \omega t$ (5 V, 9 kHz) is applied between the tip and the bottom electrode (Pt) of the sample. The nanoscale displacements induced by the electric fields were measured by the detecting of the piezoelectric vibration of the surface of PLT films due to inverse piezoelectric effect when an alternating electric field was applied between the tip and the surface of PLT films. All scans were performed at room temperature using a conductive Au coated Si_3N_4 cantilever (Seiko) with an elastic constant of 1.4 N/m and an integrated tip of about 20 nm in diameter. Furthermore, simultaneous acquisition of signals derived from wavy surface of films allows us to compare topographic and piezoelectric information.

III. RESULTS AND DISCUSSION

A. X-ray diffraction

To optimize the preparation conditions, PLT thin films were firstly deposited on $SiO_2/Si(100)$ substrates at 300 °C and then annealed via CTA at a heating rate of 3 °C/min from room temperature to 550, 575, 600, and 625 °C for 1 h, respectively. Figure 1 shows the XRD patterns of PLT thin films on $SiO_2/Si(100)$ substrates as deposited and annealed at 550, 575, 600, and 625 °C for 1 h, respectively. It can be seen clearly from Fig. 1 that the PLT thin film sputtered at 300 °C is amorphous. After annealing at 550 and 575 °C for 1 h, there is still some content of pyrochlore phase in the film, although the intensity of the pyrochlore phase (444) in the PLT film annealed at 575 °C for 1 h is remarkably weak.

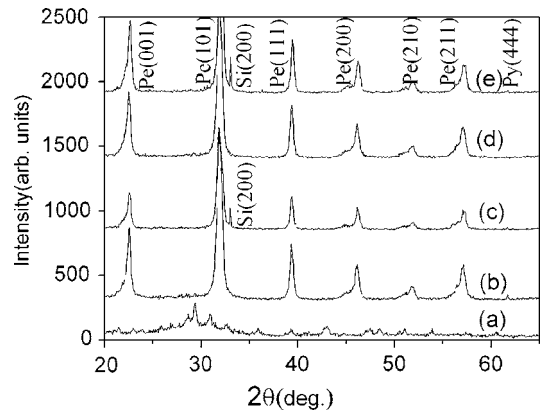

FIG. 1. XRD patterns of PLT10 films on $SiO_2/Si(100)$ substrates: (a) as deposited and annealed at (b) 550 °C, (c) 575 °C, (d) 600 °C, and (e) 625 °C, respectively, for 1 h. Pe: perovskite phase and Py: pyrochlore phase

The pure perovskite phase could be obtained and the pyrochlore phase completely disappeared when the PLT thin films annealed at 600 °C for 1 h. However, the pyrochlore phase appears again in films annealed at 625 °C due to the volatilization loss of lead. The PLT thin films deposited on $Pt/Ti/SiO_2/Si(100)$ substrates sputtered at 300 °C were also amorphous, hence they need to be annealed at certain temperature to become crystalline. According to the analysis of Fig. 1, the optimal annealing condition of PLT thin films sputtered at 300 °C on $Pt/Ti/SiO_2/Si(100)$ substrate should be 600 °C for 1 h.

Figure 2 shows the XRD pattern of the PLT thin film on $Pt/Ti/SiO_2/Si(100)$ substrate annealed at 600 °C for 1 h. Figure 2(b) shows the XRD patterns with 2θ ranging from 38° to 42°, which allows one to clearly separate the diffraction peaks of perovskite phase (111) of the PLT film from that of the Pt (111). There are only four diffraction peaks: (111) and (222) of PLT film, (111) of Pt film, and an undetermined peak, which may be a nonperovskite phase, respectively. Figure 2(b) clearly shows that perovskite structure with (111) orientation forms in the PLT films. The preferred orientation may be primarily attributable to the Pt electrode which is also (111) orientated, though the occurrence of (111) preferred orientation of tetragonal perovskite phase in PLT film depends on the deposition and annealing condi-

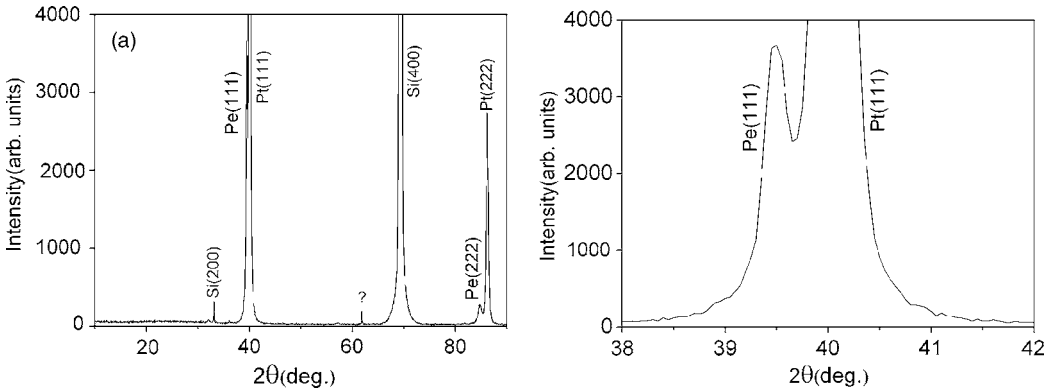

FIG. 2. (a) XRD patterns of the PLT film on $Pt/Ti/SiO_2/Si(100)$ substrate annealed at 600 °C for 1 h. (b) XRD patterns with 2θ ranging from 38° to 42° which allows one to clearly separate the diffraction peak of perovskite phase (111) of PLT film from that of Pt (111). Pe: perovskite phase.

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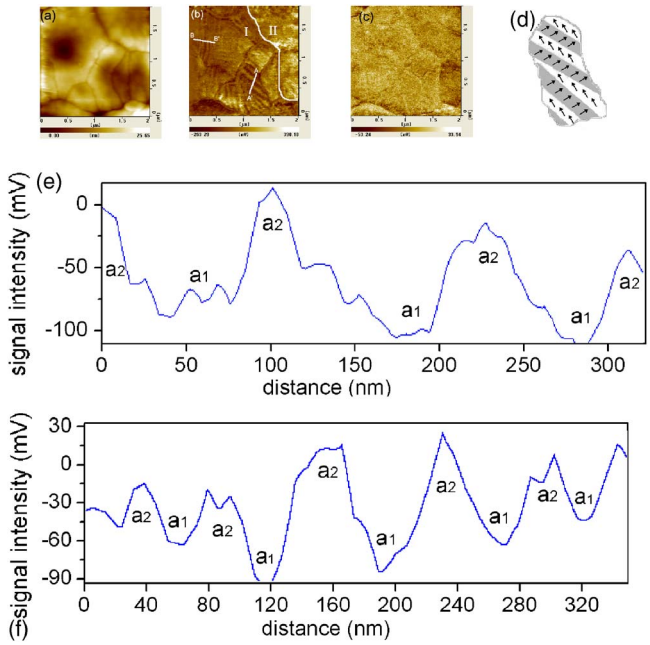

FIG. 3. (Color online) The PFM images of PLT thin film: (a) the topography image, (b) the corresponding piezoresponse image of in-plane polarization (IPP), (c) the corresponding piezoresponse image of out-of-plane polarization (OPP), (d) suggested domain pattern for the observed IPP round line *A*-*A*′ in (b), (e) line scan piezoresponse signal along line *A*-*A*′ in (b), and (f) line scan piezoresponse signal along line *B*-*B*′ in (b).

tions. Furthermore, a very small amount of nonperovskite phase still coexists with the dominant perovskite phase in the PLT film.

B. The topography and domain patterns

Figures 3(a)–3(c) show the simultaneously acquired topography image and the corresponding piezoresponse images of in-plane polarization (IPP) and out-of-plane polarization (OPP) of 2×2 μm^2 area of the PLT film, respectively.

The topography image (atomic force microscopy image in contact mode) in Fig. 3(a) shows that the root mean square (rms) roughness of the film is about 4 nm, indicating that the PLT thin films obtained at optimal preparation conditions have rather smooth and dense surface.

As shown in Figs. 3(b) and 3(c), there is scarcely any OPP in the PLT film as indicated by the approximate zero piezoresponse signal in Fig. 3(c), but banded domain patterns can be clearly seen in Fig. 3(b). Figure 3(d) illustrates the suggested domain patterns for the observed IPP around line *A*-*A*′ in Fig. 3(b). Most of these banded domains can be attributed to *a*-*a* 90° domain patterns with the polarization vector lying in the *x*-*y* plane from the mapped IPP and geometrical considerations.[8,9]

Figure 3(b) also clearly shows that parallel stripes alternated between brightness and darkness appears in most of the grains. The bright or dark lamellae correspond to the a_1 or a_2 domain,[8] as schematically drawn in Fig. 4, respectively.

The thickness of lamellae domains can be easily measured by means of line scan piezoresponse signal along lines *A*-*A*′ and *B*-*B*′ in Fig. 3(b) crossing the parallel stripes, corresponding to the hills and valleys shown in Figs. 3(e) and 3(f), respectively. It is indicated from Figs. 3(e) and 3(f) that

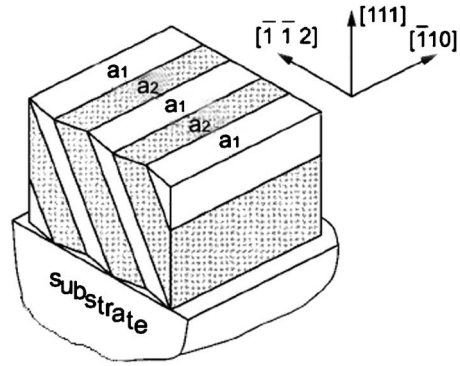

FIG. 4. Schematic drawing of 90° *a*-*a* domain structure of PLT thin film.

a-*a* domain slices were about 1 μm in height and 30 and 60 nm in width. The thicknesses of the domain slices are not the same, because they can be correlated with stress conditions.[10]

C. Stress and strain

The domain patterns in a ferroelectric film occur as the result of phase transition from the paraelectric to ferroelectric state to minimize the elastic energies of the system at the expense of creating domain boundaries and developing nonuniform elastic fields near the film/substrate interface. It is generally deemed that the domains occur in alternating patterns to relieve stresses associated with constrained transformation strains and previously unrelieved strains in the high temperature phase.[8] That is to say that the formation of alternating banded domain with adjacent lamellae is induced by the mechanical strain at the surface and at the interface between the film and the substrate since such a domain structure can minimize the mechanical energy of the film and therefore be more stable.[9] It can be seen from Fig. 4 that there exist shear strains in the interface between adjacent 90° domains.[6,8] The inner stresses in films can be divided in two classes: the intrinsic inner stress and the thermal inner stress. The intrinsic inner stress results from the compressive stresses caused by continuous injection of ions with relatively high energy into the film surface and the tensile stresses caused by micropores in film trying to shrink their surfaces. The thermal inner stress mainly comes from different thermal expansion coefficients between the film and the substrate. It can also be introduced in the processing of CTA due to nonuniform heat transmission.

It is noted that Fig. 3(b) can be separated into two regions (region I and region II), white curved line. Compared with region II which is a very small part of the whole area, region I is full of parallel slices. It can be seen clearly that the white curved line goes along grain boundaries which is also the boundaries of stresses. Based on the experiments in Ref. 11, we conclude that the different domain patterns in these two regions highlight the different stress fields in region I and region II. In addition, Figs. 3(e) and 3(f) indicate that the domain slices around line *B*-*B*′ in Fig. 3(b) are thinner and denser than those around line *A*-*A*′ due to the different stress fields. The nonuniform stress field developed inside each grain involves local deformations. The domain

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reorganization favors the local deformation due to the high rigidity of PLT films. These rearrangements are more efficient as the number of domains is increased, thus the domain size is reduced. In our experimental results, the width of lamellae domain of PLT thin film varies between 30 and 60 nm. The fragmentation of ferroelectric domains allows the minimization of the mechanical energy of the material during domain forming.[11]

Normally, the polarization vectors adopt a head-to-tail arrangement across the boundary to minimize the amount of the space charges,[12] as shown in Fig. 3(d). Most researchers believe that 90° domain arrangements follow head-to-tail rules to reduce electrical energy.[13] It is usually believed that the 90° domains may not predominate if no long-range elastic energy is taken into account, since the head-to-tail alignment of dipoles along the 90° domain wall can release the as-induced elastic energy more effectively than the 180° domains. In fact, the recent *ab initio* calculations claimed that the energy of 90° domain wall was much lower than that of 180° domain wall in bulk $PbTiO_3$.[14] According to the schematic polarization configurations shown in Fig. 3(d), it can be found that head-to-tail polarization configurations often exist in PLT thin films, indicating that the low electrical energy configurations are quite common in PLT thin films.

IV. CONCLUSIONS

PLT thin films were prepared on $Pt/Ti/SiO_2/Si(100)$ substrates by using rf magnetron sputtering. The optimal preparation conditions were to deposit at 300 °C and sequentially to anneal at 600 °C for 1 h. XRD characterizations indicated that the films were (111) oriented tetragonal perovskite phase. PFM imaging shows the existence of nanoscale banded *a*-*a* 90° ferroelectric domain patterns with lamellae of 30–60 nm in thickness in these films. Alternating banded domain with adjacent lamellae were observed and attributed to the mechanical strains at the surface and at the interface between film and substrate. Head-to-tail polarization configurations were also often observed, indicating that the low electrical energy configurations are quite common in PLT thin films.

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[1] S. J. Lee, A. K. Wang, Y. Kang, S. K. Han, M. S. Jang, B. G. Chae, Y. S. Yang, and S. H. Kim, Appl. Phys. Lett. **72**, 299 (1998).
[2] A. V. Bune, V. M. Fridkin, S. Ducharme, L. M. Blinov, S. P. Palto, A. V. Sorokin, S. G. Yudin, and A. Zlatkin, Nature (London) **391**, 874 (1998).
[3] Z. G. Wu and R. E. Cohen, Phys. Rev. Lett. **95**, 037601 (2005).
[4] W. H. Chan, Z. K. Xu, J. W. Zhai, and H. Chen, Appl. Phys. Lett. **87**, 192904 (2005).
[5] W. G. Liu, Q. Kang, L. Y. Zhang, and X. Yao, Ferroelectrics **154**, 313 (1994).
[6] C. E. Zybill *et al.*, Surf. Sci. **440**, 221 (1999).
[7] M. Labardi, C. Polop, V. Likodimos, L. Pardi, M. Allegrini, E. Vasco, and C. Zaldo, Appl. Phys. Lett. **83**, 2028 (2003).
[8] A. E. Romanov, A. Vojta, W. Pompe, M. J. Lefevre, and J. S. Speck, Phys. Status Solidi A **172**, 225 (1999).
[9] C. Loppacher, F. Schlaphof, S. Schneider, U. Zerweck, S. Grafstrom, L. M. Eng, A. Roelofs, and R. Waser, Surf. Sci. **532–535,** 483 (2003).
[10] F. Z. He, B. O. Wells, and S. M. Shapiro, Phys. Rev. Lett. **94**, 176101 (2005).
[11] P. Sarrazin, B. Thierry, and J. C. Niepce, J. Eur. Ceram. Soc. **15**, 623 (1995).
[12] F. Fang and W. Yang, Mater. Lett. **57**, 198 (2002).
[13] C. C. Chou, C. S. Hou, and T. H. Yeh, J. Eur. Ceram. Soc. **25**, 2505 (2005).
[14] B. Meyer and D. Vanderbilt, Phys. Rev. B **65**, 104111 (2002).

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